

Project Proposal

Factors Affecting Mental Health Outcomes

Project Title:

Factors Affecting Mental Health Outcomes

Purpose and Outcome:

- **Purpose:**
 - To analyze global mental health data using Python in order to examine how demographic, clinical, lifestyle, and treatment-related factors are associated with mental health outcomes.
- **Outcome:**
 - Clear analytical results illustrating the relationships between factors and mental health outcomes.
 - Identify the factors that strongly impact to positive and negative mental health outcomes.
 - Present the chart and provide relevant comments.

Dataset:

- **Source:**

<https://www.kaggle.com/datasets/ankushnarwade/global-mental-health-dataset-2025/data>
- **Description:**
 - The dataset contains 2500 records and 15 columns
 - The last updated time of the dataset: 18/12/2025
 - Each record represents an anonymous patient.
- **Structure:**
 - **Patient_ID:** Unique anonymized identifier assigned to each patient record
 - **Age:** Age of the individual in years (18 to 80)
 - **Gender:** Gender of the patient
 - **Country:** Country of residence of the patient
 - **Depression_Score:** Depression severity score based on the PHQ-9 scale (0–27)
 - **Anxiety_Score:** Anxiety severity score based on the GAD-7 scale (0–21)
 - **Stress_Level:** Self-reported stress level (Low, Medium, High, Severe)

- **Sleep_Hours :** Average number of hours slept per night
- **Physical_Activity:** Level of regular physical activity (None, Low, Moderate, High)
- **Chronic_Illness:** Indicates presence of any ongoing physical health condition (Yes/No)
- **Mental_Health_History:** Whether the patient has a prior mental health diagnosis (Yes/No)
- **Treatment:** Type of mental health treatment received (None, Therapy, Medication, Both)
- **Days_of_Treatment:** Total number of days the patient has undergone treatment
- **Outcome:** Overall treatment outcome or mental health status (Poor, Fair, Good, Excellent)
- **Work_Status:** Current employment or academic status of the patient

Initial Analysis Plan:

- **Data Cleaning:** Check and handle missing values, duplicates, correct data types, check variables have been suitable range, handle inconsistent data.
- **EDA:** Generate summary statistics, visualize distributions of mental health outcome, analyze relationships between demographic, clinical, lifestyle, and treatment factors with mental health outcomes using boxplot, bar chart. Explore correlations among numeric variables and identify patterns that may influence outcomes.
- **Analysis:** Build ordinal logistic regression to evaluate which factors most strongly predict Outcome.
- **Visualization:** Generate correlation heatmaps for numeric variables
- **Data Storytelling:** Present the analysis results with clear visuals and insights. Identify the most influential factors affecting mental health outcomes. Provide actionable recommendations to improve mental health care and well-being.